

Pooja Kiran

Security Engineer | AI & Agent Security | Application Security | Cloud IAM Security | Model Supply-Chain Security
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EXPERIENCE

Independent AI Security Engineer

Aug. 2024 - Present

Self-Directed Security Projects

Tempe, AZ, USA

- Mapped agent-to-tool, cloud IAM, and model supply-chain threats to enforceable controls, static checks, and repeatable regression tests.
- Automated Python regression and GitHub Actions validation across 3 self-directed security projects; dated CI evidence includes **723 gateway tests**, **25 IAM rule IDs**, and **241 model-scanner tests**.

Business & Compliance Lead, AEROSSEC (Academic Project)

Aug. 2025 - Dec. 2025

ASU & Honeywell Aerospace Technology Innovation Lab (TEM 598)

Tempe, AZ, USA

- Developed compliance and commercialization analysis for a proposed airline PSS security layer, including a **\$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

PROJECTS

Project periods are candidate-reported; metrics cite dated 2026 CI snapshots.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security*

Oct. 2025 - Sep. 2026

- Built a default-deny MCP/JSON-RPC 2.0 gateway with 5 pre-execution controls: capability authorization, prompt-injection inspection, PII/exfiltration checks, anti-SSRF egress policy, and rate limiting.
- Expanded the prompt-injection collection from **55 earlier patterns to 69 current patterns**, with tamper-evident policy-decision audit logging.
- Cited CI snapshot: **723 passing tests**; a separate snapshot measured **81.91% statement coverage**. Includes **9 Elastic rules** and **21 core SIEM tests**.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0*

Apr. 2025 - Sep. 2025

- Built an IAM static analyzer with **25 deterministic rules** covering wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy weaknesses, audit tampering, and permission boundaries.
- Produced 3 finding formats (text, JSON, SARIF 2.1.0), integrating SARIF with GitHub Code Scanning for reviewable IAM policy risks.
- Verified CI snapshot: **243 collected, 240 passed, 3 skipped**; skipped checks require live AWS credentials.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras*

Nov. 2024 - Mar. 2025

- Engineered non-executing inspection across 5 model artifact families: SafeTensors, GGUF, ONNX, Keras, and pickle-derived files; avoided running untrusted model code by default.
- Combined pickle-opcode, AST/taint, provenance/dependency, and obfuscation checks; detected 33/33 committed adversarial fixtures (12 incident recreations, 18 variants, 3 large-scale cases).
- Verified CI snapshot: **241 passed, 1 skipped, 6 additional pytest subtests passed** with **75.67% statement coverage**; fixture detection is evaluated separately.

EDUCATION

Arizona State University

Tempe, AZ, USA

M.S., Information Technology (Security), GPA: 3.87/4.00

Aug. 2024 - May 2026

- **Academic Participation:** Student participant, ASU, 2025; Generative AI learning assistant for personalized education.

M. S. Ramaiah University of Applied Sciences

Bengaluru, KA, India

B.Tech., Computer Science & Engineering, CGPA: 8.44/10

Aug. 2019 - Aug. 2023

TECHNICAL SKILLS

AI & Agent Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security

Identity & Application Security: Cloud IAM Security, AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Capability Authorization, Threat Modeling, API Security

Security Engineering & DevSecOps: Python, FastAPI, pytest, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, Docker

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Tamper-Evident Audit Logging, MITRE ATLAS

CERTIFICATIONS

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)